

University of Glasgow

Classification of Hypertension Subtypes from Circulating microRNA Profiles Using Supervised Machine Learning under Class Imbalance

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Summary

Hypertension is a widespread condition that significantly increases the risk of cardiovascular disease. Most cases are classified as primary hypertension (PHT). Over 10% are attributed to endocrine hypertension (EHT), which results from hormonal disorders such as primary aldosteronism (PA), Cushing's syndrome (CS), or tumors like pheochromocytoma and paraganglioma (PPGL). These forms of EHT are often mistakenly treated as PHT, leading to delays in diagnosis, prolonged use of ineffective medications, and increased burden on patients' health. Diagnosing EHT is clinically challenging because each subtype requires distinct protocols, such as pausing medications, fasting, early-morning sample collection, or specialized imaging and hormonal tests, which are inconvenient for patients.

Circulating microRNAs have recently emerged as promising non-invasive biomarkers, with specific profiles linked to hypertension subtypes. At the same time, machine learning (ML) methods have gained momentum in biomarker discovery, particularly for large-scale datasets from next-generation sequencing. In this study, retrospective plasma microRNA samples (n= 36 (CS), 109 (PA), 111 (PHT), 76 (PPGL), 133 (HV)) from the European ENS@T-HT Horizon 2020 consortium were classified to investigate whether ML could be effectively applied to identify hypertension subtypes. A major challenge was class imbalance, with some subgroups being considerably smaller than others, which can bias models toward majority classes.

To address this, evaluation metrics better suited to imbalanced data were applied. Balanced accuracy replaced standard accuracy, and a novel measure, minimum sensitivity, was introduced. Sensitivity was first calculated for each subgroup, and the lowest value among them, representing the worst-performing subclass, was defined as the minimum sensitivity to ensure that rare subtypes were not overlooked. For example, a model with high balanced accuracy but poor sensitivity for the worst-performing subgroup, CS, would risk many missed diagnoses. Thus, models achieving higher minimum sensitivity were prioritized. Four models, Random Forest, XGBoost, CatBoost, and Support Vector Machines (SVM), were implemented and optimized for minimum sensitivity through class-weight allocation for minority groups. Hyperparameters were tuned using exhaustive and adaptive grid searches. The best-performing hyperparameter sets were subsequently combined through ensemble

stacking, in which predictions from the four models were integrated to make final classifications.

Minimum sensitivity increased from 0.00 under default settings to 0.42 after tuning with SVM. Overall balanced accuracy reached 0.65 with stacking, indicating moderate average performance across subtypes. One microRNA was found to strongly contribute to classification of the CS group, highlighting potential biological relevance. Notably, sensitivity for the smallest subgroup, CS, increased from 0.00 to 0.71 through optimisations.

The study demonstrated that optimizing ML models and applying fair evaluation metrics can substantially improve classification of small subgroups in imbalanced data. Although current performance remains insufficient for clinical application, the findings provide proof-of-concept that microRNA profiles combined with ML could hold promise for non-invasive hypertension subtype diagnosis. Future studies with larger cohorts and multi-omics integration may further enhance clinical utility.

1.Introduction

Hypertension is a leading global health problem, affecting approximately 20% to 50% of adults in Europe and substantially increasing the risk of cardiovascular morbidity[1]. While the majority of cases are classified as primary hypertension (PHT), an important minority is attributed to endocrine hypertension (EHT), which arises from hormonal disorders such as primary aldosteronism (PA), Cushing's syndrome (CS), and catecholamine-secreting tumors including pheochromocytoma and paraganglioma (PPGL)[2]. Estimates suggest that at least 10% of hypertensive patients fall into the category of EHT, with some sources regarding this figure as an underestimate of the true prevalence[3].

The distinction between PHT and EHT is clinically important, as management strategies differ fundamentally. PHT is generally treated with antihypertensive agents, whereas EHT requires medications that regulate hormone levels or surgical interventions[2,3]. However, EHT is frequently misdiagnosed as PHT owing to overlapping symptoms and the complexity of diagnostic workflows. Consequently, patients are often subjected to inappropriate treatment, delayed diagnosis, and prolonged morbidity[2,4]. Diagnosis of EHT remains challenging. Each subtype requires distinct, often cumbersome, diagnostic protocols, some necessitate pausing of medications or fasting, others require early-morning sampling, and several rely on advanced imaging or specialized hormonal testing. No single diagnostic test is currently available to reliably screen for all forms of EHT[2]. This limitation has prompted interest in novel, non-invasive biomarkers.

Circulating microRNAs have emerged as promising candidates. These small non-coding RNAs regulate gene expression and are stable in plasma, making them suitable for non-invasive testing[5]. Distinct microRNA profiles have been linked to different pathological states, including hypertension subtypes[5,6]. In parallel, machine learning (ML) methods have expanded rapidly in biomedical research. ML algorithms are capable of detecting complex, non-linear patterns in high-dimensional datasets and are increasingly applied to diagnostic problems where conventional methods are limited[7-9].

In this study, retrospective microRNA samples from the European ENS@T-HT Horizon 2020 consortium were analyzed using supervised ML models. The ENS@T-HT consortium

investigates hypertension with the aim of developing more personalised approaches to treatment and management[10]. The dataset comprised samples from patients with CS ($n = 36$), PA ($n = 109$), PHT ($n = 111$), PPGL ($n = 76$), and healthy volunteers (HV, $n = 133$). A major challenge in applying ML to such clinical datasets is class imbalance, whereby certain subgroups are relatively well represented, while others, often of high clinical relevance, are rare. Under these conditions, conventional ML algorithms tend to favor majority classes, yielding poor performance in minority groups such as CS[6,11-13].

Another limitation arises from the evaluation of ML models using conventional performance metrics. Standard accuracy can be misleading in the context of imbalanced data, as high overall accuracy may be achieved by consistently predicting majority classes while failing to detect rarer subtypes[12]. To address this issue, balanced accuracy was employed, as it accounts for per-class performance by averaging sensitivity across groups[13]. In addition, minimum sensitivity was introduced as a novel fairness-oriented metric. Rather than averaging sensitivity across all subtypes, the sensitivity of each individual subgroup was evaluated, and the lowest value among them was defined as the minimum sensitivity. This approach ensures that the performance of the worst-performing class is explicitly considered, thereby reducing the risk of rare groups such as CS or PPGL being overlooked.

Against this background, the central research question of the present study was formulated: *Can supervised machine learning methods applied to circulating microRNA profiles accurately classify hypertension subtypes despite challenges posed by small sample size and class imbalance?*

To address this question, the study was designed with the following objectives:

1. To address class imbalance by applying fair evaluation metrics (balanced accuracy and minimum sensitivity) and by adjusting class weights during training.
2. To optimise model performance through hyperparameter tuning using exhaustive and adaptive grid search.
3. To explore ensemble learning through stacking to improve robustness.
4. To investigate the contribution of individual microRNAs to subtype classification and identify potential biomarker candidates.

Four supervised ML algorithms were systematically evaluated and optimized for minimum sensitivity: Random Forest, XGBoost, CatBoost, and Support Vector Machines (SVM). Class imbalance was addressed by applying higher weights to minority classes during training. Hyperparameter optimization was performed using both exhaustive and adaptive grid searches, and stacking was applied to integrate predictions into an ensemble classifier. These strategies markedly improved sensitivity in rare subtypes such as CS, while also identifying individual microRNA features with strong discriminatory potential. Although overall balanced accuracy did not substantially increase after optimization, minimum sensitivity was significantly enhanced, demonstrating improved fairness across subtypes.

Taken together, these findings show that while circulating microRNA profiles alone are not yet sufficient for reliable clinical application, optimization of ML models and the use of fairness-oriented evaluation metrics can substantially improve classification of minority subgroups in imbalanced clinical data. The results provide proof-of-concept that combining microRNA profiling with advanced ML approaches offers promise as a non-invasive diagnostic strategy for hypertension subtypes.

2. Materials and Methods

2.1 Study Cohort and Sample Collection

Retrospective EDTA-plasma samples from 465 participants diagnosed with CS, PA, PHT, PPGL, and healthy volunteers (HV) were obtained from the ENS@T-HT Horizon 2020 project archives (Table 1). All samples were collected during previous clinical studies conducted in 11 collaborating centres. Ethical approval, participant recruitment criteria, and detailed inclusion and exclusion protocols are described in Reel et al. (2025)[6], which analyzed an overlapping subset of the same cohort.

Total RNA was extracted from 200 μ L of EDTA-plasma using the miRNeasy Mini Kit (QIAGEN, Manchester, UK). MicroRNA expression profiling was performed on Serum/Plasma Focus microRNA PCR Panels (384-well, V4.M, Exiqon) in combination with ExiLENT SYBR® Green master mix and analyzed on a QuantStudio 12K Flex Real-Time PCR System (Thermo Fisher). Full details of cDNA synthesis, quality control, and laboratory workflows are provided in Reel et al. (2025)[6].

Table 1. Distribution of samples across hypertension subtypes and healthy volunteers

Subgroup	Number of samples (<i>n</i>)
Cushing's syndrome (CS)	36
Primary aldosteronism (PA)	109
Primary hypertension (PHT)	111
Pheochromocytoma/Paraganglioma (PPGL)	76
Healthy volunteers (HV)	133
Total	465

2.2 MicroRNA Quantification and Data Preprocessing

Raw cycle threshold (Ct) values generated by qRT-PCR were processed using GenEx software (v6, MultiD Analyses, Vedbaek, Denmark). Cross-plate variability was corrected through interplate calibration using UniSp3 spike-in controls, ensuring consistent normalization across runs.

The final dataset consisted of ΔCt (DeltaCt) values, calculated by normalizing each Ct value to the mean of the five most stable reference microRNAs within its plate. Because Ct represents the PCR cycle at which fluorescence exceeds the detection threshold, lower Ct values indicate higher microRNA abundance. Consequently, smaller ΔCt values correspond to greater relative expression, whereas larger ΔCt values reflect lower expression levels. The qRT-PCR scale is logarithmic, such that a one-cycle difference approximates a two-fold change in expression.

For microRNAs where amplification failed in specific samples, missing ΔCt values were imputed by assigning a surrogate Ct equal to one cycle higher (i.e. half the expression) than the highest reliably measured Ct (lowest reliable expression) for that microRNA. This approach preserves the relative expression ranking without introducing artificial variability. Spike-in controls (UniSp) and a synthetic negative control (cel-) were retained for quality monitoring but excluded from downstream machine learning analyses[6].

All sample and microRNA identifiers were anonymized to prevent analyst bias.

2.3 Exploratory Feature Adjustments

Exploratory adjustments were evaluated to determine whether they could improve model performance.

2.3.1 Covariate Regression

MicroRNA ΔCt values were adjusted for potential confounding effects of sex and age using ordinary least squares (OLS) regression. Sex and age are recognized sources of biological variability in circulating microRNA expression, as both hormonal status and aging processes influence gene regulation and biomarker profiles[14]. Samples lacking sex or age information were excluded prior to regression to ensure reliable adjustment. The residuals from the OLS model were retained as covariate-corrected expression values for downstream analyses. Model performance was subsequently compared between classifiers trained on covariate-adjusted and unadjusted data.

2.3.2 Binary Detection Flags

Binary detection flags were generated to indicate whether each measurement was undetected (flag = 1) or successfully detected (flag = 0). “Undetected” was defined as having a ΔCt value one cycle higher than the highest reliably measured ΔCt for a given microRNA, corresponding to failed amplification and very low expression. Models trained with these detection flags as additional features were compared against those using ΔCt values alone.

2.4 Software and Computational Environment

All analyses were performed in Python v3.12.11 within Google Colab environments. The following key libraries were used:

- scikit-learn v1.6.1 - machine learning model implementation (including GridSearchCV, RandomForestClassifier, and SVM), cross-validation, and performance metrics[15]
- Optuna v4.5.0 - Bayesian hyperparameter optimization[16]
- XGBoost v3.0.4 and CatBoost v1.2.8 - gradient-boosted tree classifiers[17,18]
- statsmodels v0.14.5 - covariate regression adjustments using OLS[19]
- NumPy v2.0.2 and pandas v2.2.2 - data preprocessing and management[20,21]
- Matplotlib v3.10.0 and Seaborn v0.13.2 - data visualization and plotting[22,23]
- SHAP v0.48.0 - model interpretability and feature importance analysis[24]

Model training and optimization were executed on the University of Glasgow’s MVLS Advanced Research System (MARS) high-performance computing (HPC) cluster, using 8 CPU cores per task. To ensure reproducibility, all random seeds were fixed (`random_state = 42`) across preprocessing, model training, and evaluation.

2.5 Model Training and Hyperparameter Optimization (GridSearchCV)

Models were trained on two versions of the ΔCt dataset throughout the study:

- Full dataset - comprising all four patient groups and healthy volunteers
- Patient-only dataset - excluding healthy controls

The full dataset and the patient-only dataset were both derived from the original ΔC_t dataset, without sex- or age-based regression adjustments or feature engineering, and were used for model training throughout the study.

Four supervised machine learning classifiers were implemented: Random Forest, an ensemble of decision trees trained with bootstrap aggregation that reduces variance and improves stability[25]; XGBoost, a gradient-boosted decision tree algorithm optimized for speed and regularization[17]; CatBoost, a gradient boosting method designed to handle categorical features efficiently and reduce overfitting[18]; and Support Vector Machines (SVM), which construct a maximal-margin hyperplane to separate classes in high-dimensional space[26]. All models were first run under default settings to establish baseline performance.

To address class imbalance across hypertension subtypes, class weighting was applied where supported. In Python's scikit-learn, this is set as `class_weight="balanced"` for `RandomForestClassifier(sklern.ensemble)` and `SVC(sklern.svm)`, which scales class weights inversely to class frequency (i.e., rarer classes receive larger weights). For XGBoost (`xgboost.XGBClassifier`), which does not provide a `class_weight` parameter for multiclass problems, the same inverse-frequency weights were applied as per-sample weights via the `sample_weight` argument to `.fit(...)`. For CatBoost (`catboost.CatBoostClassifier`), class imbalance was handled by supplying an inverse-frequency vector through the `class_weights` parameter (ordered by class label). These settings ensured that underrepresented subtypes contributed more strongly during training across all models.

A stratified 5-fold cross-validation procedure was applied (`StratifiedKFold` from `sklearn.model_selection`), ensuring that class distributions were preserved in each fold. In each iteration, ~80% of data was used for training and 20% for validation, and results across folds were averaged to yield stable performance estimates.

Hyperparameter tuning was performed using `GridSearchCV` (`sklearn.model_selection`). Hyperparameters are model configuration parameters that are not learned directly from the data (e.g., the number of trees in Random Forest,

learning rate in boosting algorithms, or kernel type in SVM). Hyperparameter tuning is the process of systematically searching across predefined parameter ranges to identify the configuration that optimizes performance. In this study, balanced accuracy was used as the optimization metric to account for unequal class distributions by equally weighting sensitivity across subtypes. Predefined hyperparameter search spaces were specified for each model and applied consistently across both datasets to allow fair comparisons. The parameter search boundaries that yielded the best-performing configurations are shown in Supplementary Figure S1, with the corresponding configurations reported in Supplementary Table S1 and retained for downstream analyses.

2.6 Model Training and Hyperparameter Optimization (Optuna)

Optuna[16] was used as an alternative hyperparameter optimization framework for the same four classifiers, following the same ΔC_t datasets, class imbalance handling, optimization objective, parameter search boundaries, and stratified 5-fold cross-validation strategy described in section 2.5.

Unlike exhaustive grid search, Optuna employs an adaptive search strategy using the Tree-structured Parzen Estimator (TPE) sampler[27] to efficiently explore promising hyperparameter regions. A maximum of 200 trials per model was allowed, but early stopping was applied: optimization halted if no improvement of at least 0.002 in balanced accuracy was observed within the 30 most recent trials.

The best-performing configurations for each model and dataset were retained for downstream comparisons with GridSearchCV results.

2.7 Model Stacking

Model stacking is an ensemble learning technique in which the predictions of multiple base classifiers are combined by a meta-learner to improve overall performance. By integrating complementary strengths of different models, stacking often achieves greater robustness and generalization than any single classifier alone[28].

In this study, the best hyperparameter configurations for Random Forest, XGBoost, CatBoost, and SVM were selected from the combined pool of results obtained from GridSearchCV and Optuna. The best configuration for each model was defined as the parameter set achieving the highest minimum sensitivity across all subtypes, prioritizing robust performance on the weakest-performing class. These four optimized models were then used as the base learners in the stacking framework.

A logistic regression model (`LogisticRegression(max_iter=1000, random_state=42)`) was employed as the meta-learner to combine predictions from the base models. Stacking was performed using a stratified 5-fold cross-validation strategy to preserve class distributions within each fold.

To address class imbalance consistently across the entire stacking framework, sample weights were propagated to all estimators via `set_config(enable_metadata_routing=True)`. While the base models already handled imbalance internally (e.g., `class_weight="balanced"`), this step ensured that the logistic regression meta-learner also incorporated sample-level weighting, preventing bias toward majority subtypes.

2.8 Identifying Key microRNAs Driving Classification via SHAP

SHAP (SHapley Additive exPlanations)[24] analysis was applied to the three tree-based models (Random Forest, XGBoost, and CatBoost) to identify the microRNAs contributing most strongly to subtype classification. SHAP provides optimized TreeExplainer[24] implementation for tree-based models, enabling efficient and accurate computation of feature contributions. Positive SHAP values indicate a shift toward a given class, whereas negative values indicate opposition. Feature importance rankings were derived by averaging absolute SHAP values across all samples, and beeswarm[24] plots were generated to visualize the distribution of feature effects.

3. Results

3.1 Dataset Demographics

The demographic distribution of the dataset across the four disease subtypes (CS, PA, PHT, PPGL) and healthy controls (HV) was examined to assess potential confounding effects of sex and age (Figure 1). Samples lacking sex or age information were excluded from these analyses, which explains the slight discrepancy between group totals here and the overall dataset size. The HV group showed a markedly skewed sex ratio, with a substantially higher proportion of males compared to females (69% male), and a noticeably younger median age of 26 years compared to the patient groups, whose median ages ranged from 46 to 55 years. These disparities suggest that the HV group may not represent a demographically comparable control cohort. In contrast, the CS group was strongly female-dominated, with 24 females and only 6 males. The other subtypes (PA, PHT, and PPGL) exhibited more balanced sex distributions and similar median ages, making them more comparable to one another and less likely to introduce demographic bias relative to HV or CS. These findings indicate that sex and age may act as confounders in classification and justify explicit adjustment in subsequent analyses.

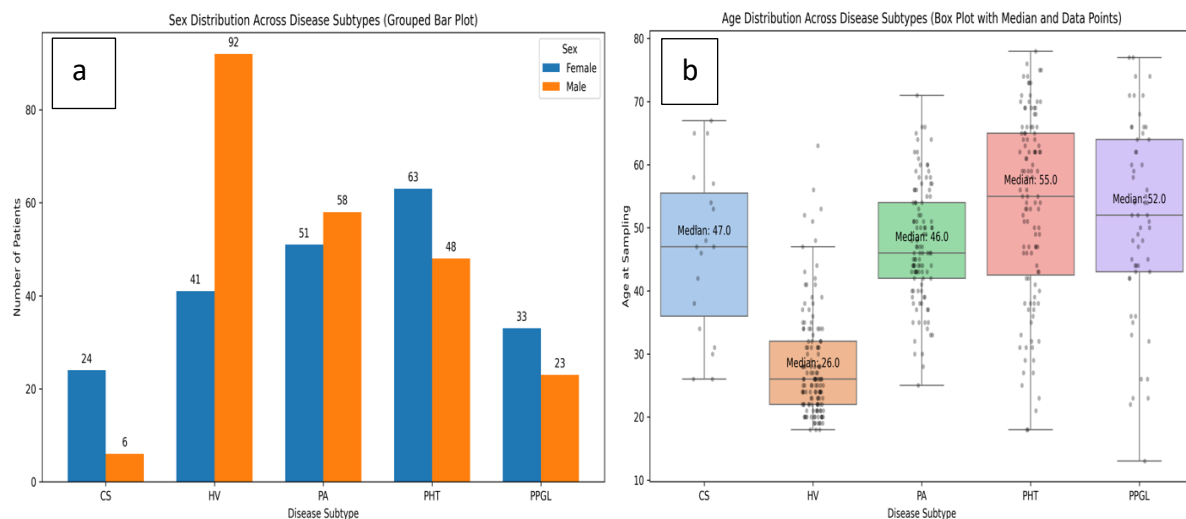


Figure 1. Demographic distribution across hypertension subtypes and healthy volunteers (HV).

(a) Grouped bar plot of sex distribution across groups, illustrating male predominance in the HV group and female predominance in the CS group.

(b) Box plot of age distribution at sampling across disease subtypes and HV, with individual data points overlaid. Median ages are indicated. The HV group was observed to have a substantially younger median age compared to patient groups, while the patient groups displayed broadly similar age distributions.

Note: Samples lacking sex or age information were excluded from these plots.

3.2 Effect of Sex/Age Regression and Feature Engineering

The impact of removing potential confounding variables and enhancing feature representation was assessed by testing two pre-processing strategies under default model settings:

- (1) regressing out sex and age, and
- (2) introducing binary detection flags to indicate whether individual miRNAs were detected in each sample (1 = undetected, 0 = detected).

When the Random Forest model was applied, regressing out sex and age reduced balanced accuracy (0.44 vs. 0.51) and lower sensitivities for most subtypes, particularly CS (0.00 vs. 0.06) and PPGL (0.10 vs. 0.40) (Fig. 2a–b). Similarly, the introduction of detection flags for feature engineering did not improve performance, leading instead to a slight reduction in balanced accuracy (0.49 vs. 0.51) and minimal or negative effects on class-wise sensitivity (Fig. 2c–d).

Random Forest was presented as a representative example, as similar patterns were observed across SVM, XGBoost, and CatBoost. Collectively, these results indicated that neither regressing out sex and age nor feature engineering enhanced classification under default model settings. Consequently, the non-regressed, non-engineered dataset was selected as the baseline for subsequent model optimization.

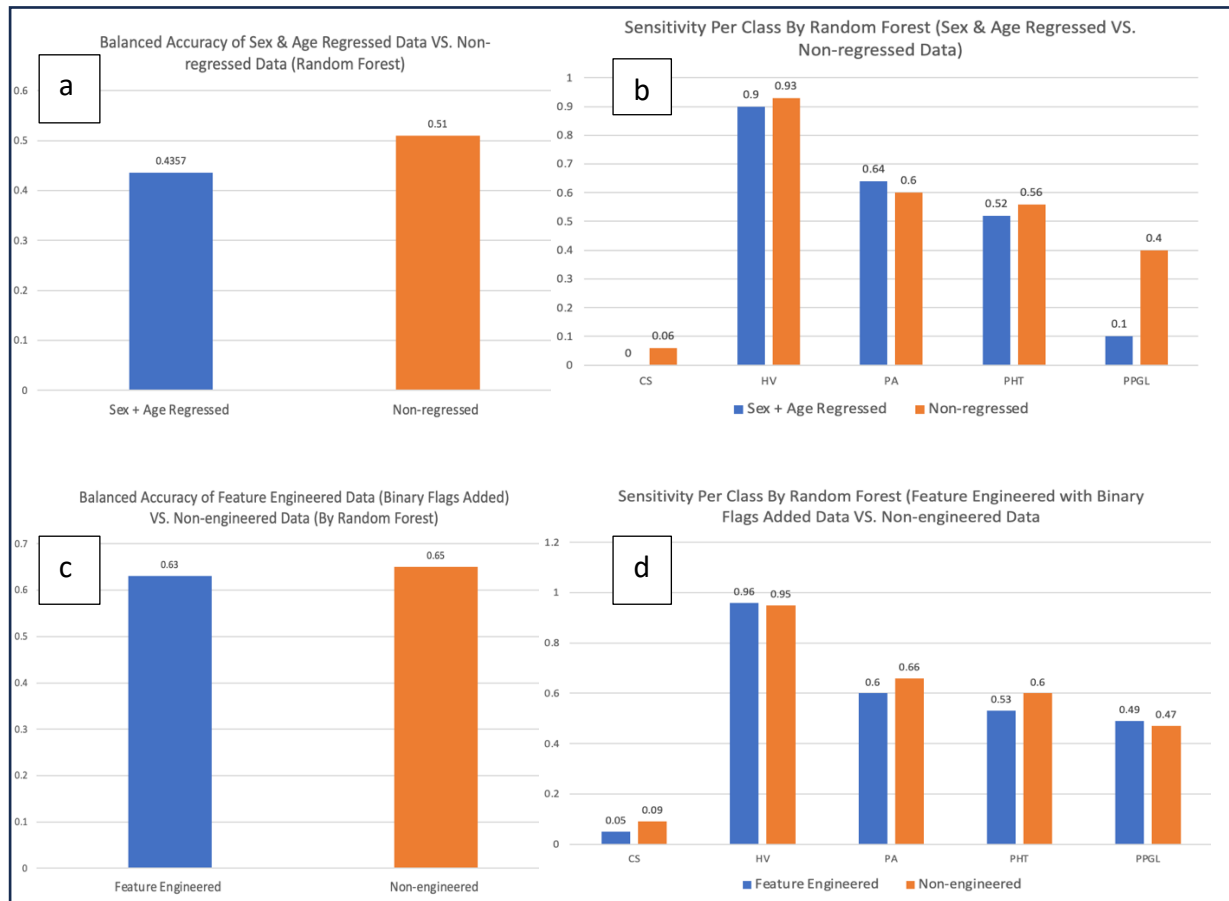


Figure 2. Impact of sex/age regression and feature engineering on Random Forest model performance.

(a) Balanced accuracy comparison between sex- and age-regressed data and non-regressed data.

(b) Sensitivity per class for regressed vs. non-regressed datasets. A slight reduction in both balanced accuracy and sensitivity across most subtypes was observed when sex and age were regressed out.

(c) Balanced accuracy comparison between feature-engineered data (binary detection flags added) and non-engineered data.

(d) Sensitivity per class for engineered vs. non-engineered data. No improvement was observed when binary detection flags were included, and sensitivity in most subtypes was slightly reduced. Results are shown for Random Forest only, as similar patterns were observed for SVM, XGBoost, and CatBoost.

3.3 Baseline Model Performance on Full vs. Patient-Only Datasets

To evaluate baseline model performance under default hyperparameter settings, four supervised machine learning algorithms, Random Forest, XGBoost, CatBoost, and SVM, were applied to two settings: (i) the full dataset (patients + HV) and (ii) the patient-only dataset (HV excluded).

Balanced accuracy (Fig. 3a) was found to be generally higher when models were trained on the full dataset compared to the patient-only dataset, except for Random Forest, which showed comparable performance across both settings (~ 0.51). However, when class-wise sensitivity was assessed (Fig. 3b-e), improved sensitivities were observed for most subtypes in the patient-only setting, particularly for the smallest but clinically important class, CS (Fig. 3b) and the biggest patient subgroup, PHT (Fig. 3d). These findings indicate that inclusion of the HV group led to inflated overall balanced accuracy but did not enhance the detection of rare subtypes.

Sensitivity profiles across models and classes (Fig. 3f-g) further compared model performance using the full dataset (Fig. 3f) versus the patient-only dataset (Fig. 3g). SVM consistently underperformed, particularly for minority classes such as CS and PPGL, where sensitivities were zero, suggesting limited model generalization for underrepresented groups. In contrast, XGBoost and CatBoost achieved the highest sensitivities across most classes, especially for PA and PHT, while Random Forest performed moderately but less effectively than the gradient-boosting methods.

Overall, these results demonstrated that exclusion of the HV group resulted in reduced balanced accuracy but improved minimum sensitivity, which was identified as a critical factor for reliable detection of rare subtypes in clinical settings. These findings justified expanding subsequent optimization and performance assessments to include evaluations on the patient-only dataset alongside the full dataset.

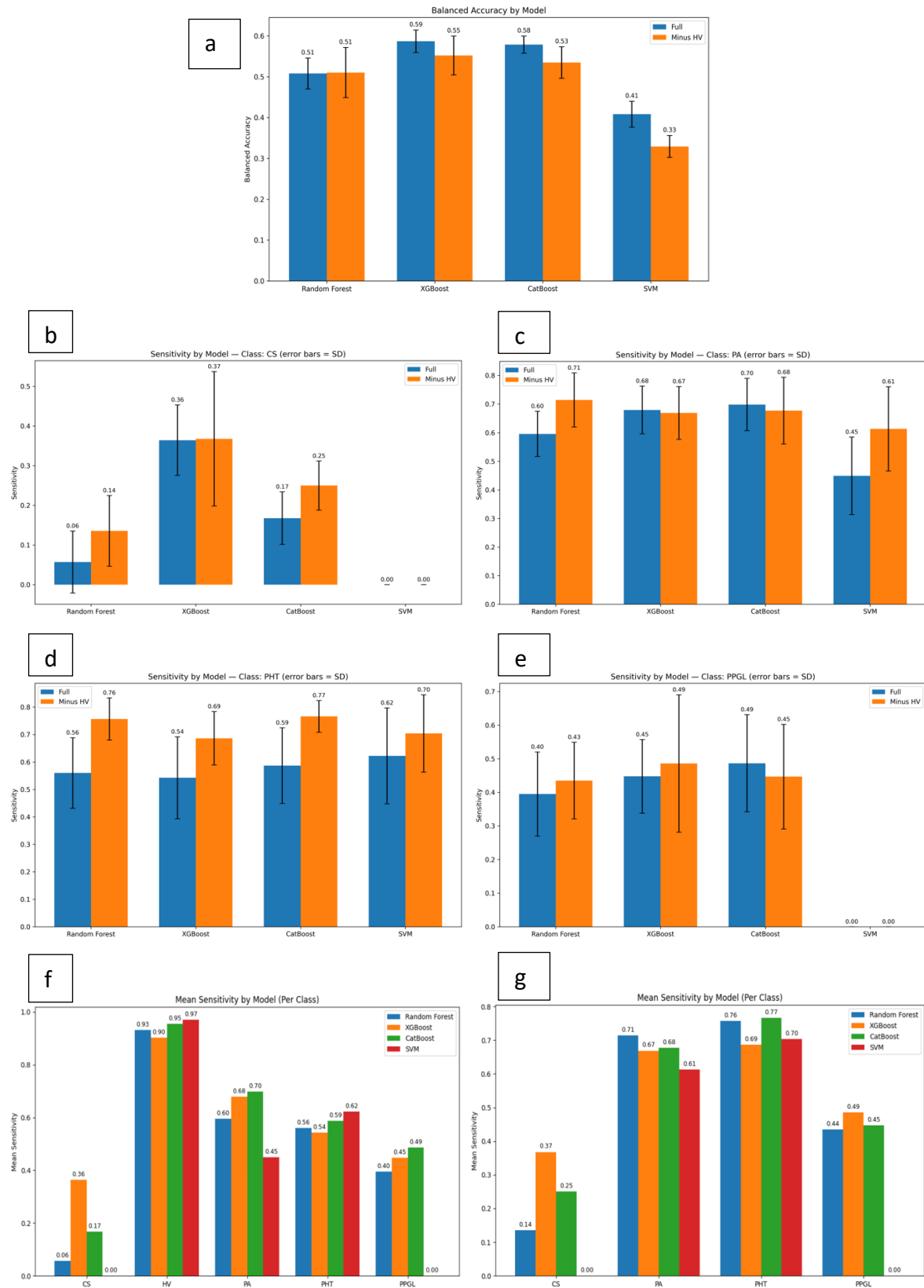


Figure 3. Baseline performance of four machine learning models under default settings using the full dataset versus the patient-only dataset (HV excluded).

(a) Balanced accuracy for Random Forest, XGBoost, CatBoost, and SVM trained on the full dataset (blue) versus the patient-

only dataset (orange). Overall balanced accuracy was slightly higher when HV samples were included, except for Random Forest, which showed comparable performance in both settings.

(b-e) Sensitivity per class ((b)CS, (c)PA, (d)PHT, (e)PPGL) comparing full versus patient-only settings across all models, where higher sensitivities for rare classes (e.g., CS, PPGL) were generally observed when HV was excluded. Error bars represent ± 1 standard deviation (SD) across cross-validation folds.

(f-g) Mean class-wise sensitivities for each model when trained on the full dataset (f) and patient-only dataset (g). SVM showed poor sensitivity for minority classes, particularly CS and PPGL. XGBoost and CatBoost achieved the best sensitivities across most subtypes, while Random Forest performed moderately.

3.4 Impact of Hyperparameter Optimization and Stacked Ensemble Modeling

To improve baseline model performance, particularly for underrepresented classes, hyperparameter tuning was first performed using two approaches: GridSearchCV and Optuna. Each of the four machine learning models, Random Forest, XGBoost, CatBoost, and SVM, was independently optimized using both methods. The best-performing configurations were selected based on minimum sensitivity. Across most models, GridSearchCV demonstrated superior performance to Optuna (Fig. 4), and the best-performing configuration from each of the four algorithms was selected for inclusion in the subsequent stacking ensemble.

These four optimized models were then used as base learners in a stacked ensemble framework, with logistic regression serving as the meta-learner. This ensemble approach was employed with the expectation that combining the complementary strengths of individual models could further enhance classification performance, particularly for rare subtypes. All evaluations were conducted on both the full dataset (patients + HV) and the patient-only dataset (HV excluded).

As shown in Figure 4a, balanced accuracy increased modestly following hyperparameter optimization, with GridSearchCV yielding slightly better performance than Optuna. The largest gain was observed for SVM, which initially performed poorly under default settings. In addition to balanced accuracy, minimum sensitivities for all four models improved substantially through tuning: Random Forest (0.06 $\rightarrow$ 0.42), XGBoost (0.36 $\rightarrow$ 0.48), CatBoost (0.17 $\rightarrow$ 0.51), and SVM (0.00 $\rightarrow$ 0.42) (Fig. 4). These improvements highlight that tuning enhanced the detection of worst-performing subtypes across all algorithms, with particularly notable gains for Random Forest and SVM. The stacked ensemble achieved the

highest balanced accuracy on the patient-only dataset (0.65), though the improvement over individual tuned models was only incremental.

3.4.1 Optimization effect on small subclasses

Class-wise sensitivity analysis revealed more substantial improvements. For the CS subgroup (Fig. 4b), sensitivity under default settings was 0 for SVM and low across the other models, reaching at best 0.37 with XGBoost in the patient-only dataset. Hyperparameter optimization with GridSearchCV or Optuna increased sensitivity to ~50% (0.42–0.56). The highest sensitivity (0.71) was achieved with stacking on the patient-only dataset, underscoring the added benefit of ensemble stacking for detecting this rare subtype. For PPGL (Fig. 4c), hyperparameter optimization produced more modest gains overall; however, SVM demonstrated a pronounced improvement (0.00 to 0.46), highlighting the importance of tuning for this model. Stacking did not yield further benefits beyond the individually optimized classifiers.

3.4.2 Optimization effect on big subclasses

For PA (Fig. 4d), sensitivity either remained similar to default settings or worsened slightly following individual optimization. A modest improvement was observed with stacking (0.77 on the patient-only dataset). The PHT group (Fig. 4e), which was the most abundant patient subclass, saw a decline in sensitivity following optimization.

These results demonstrated that hyperparameter optimization and stacking were especially effective in improving detection of rare subtypes such as CS and PPGL, with a slight worsening of performance in the largest group(s). However, these performance gains came at the cost of increased computational demands, which were further evaluated in the subsequent analysis (Fig. 5).

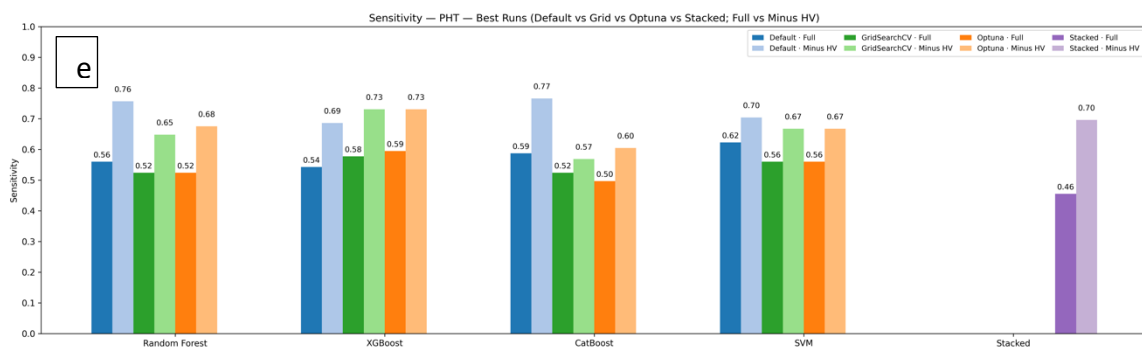
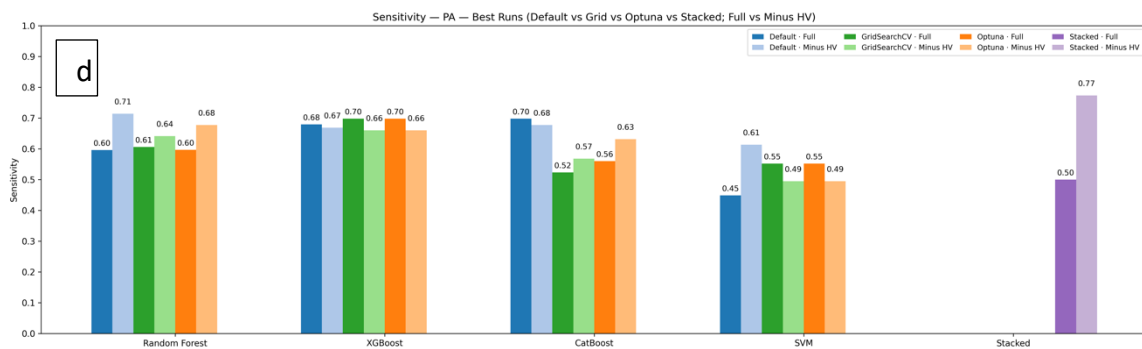
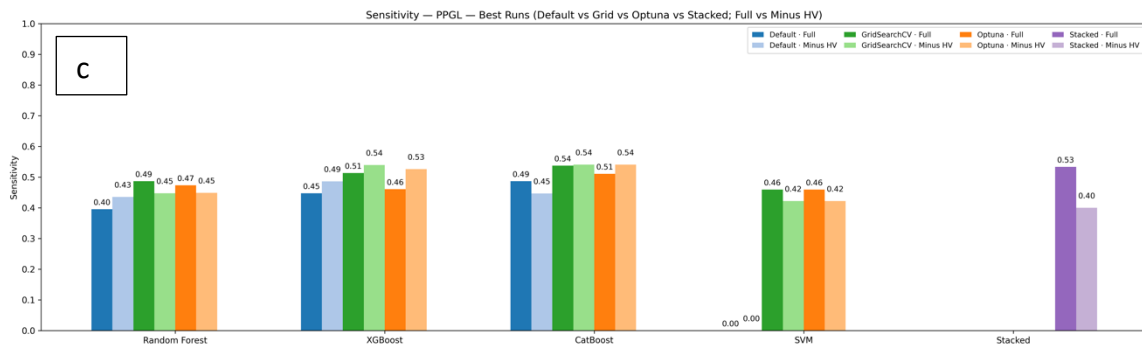
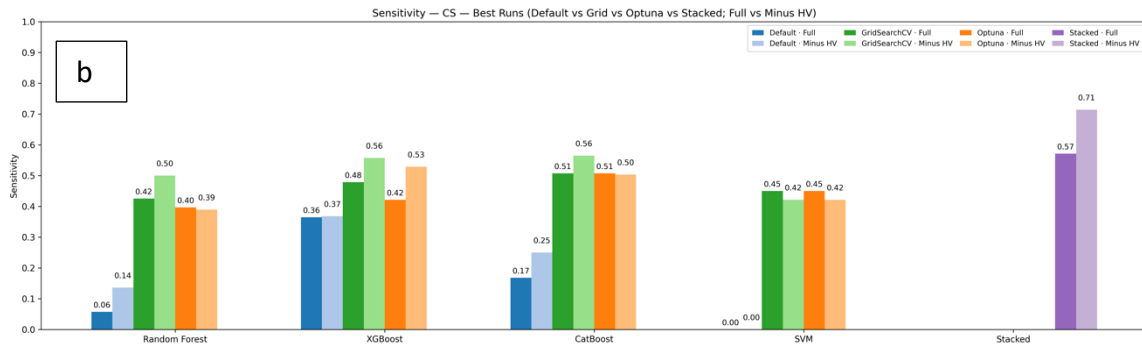
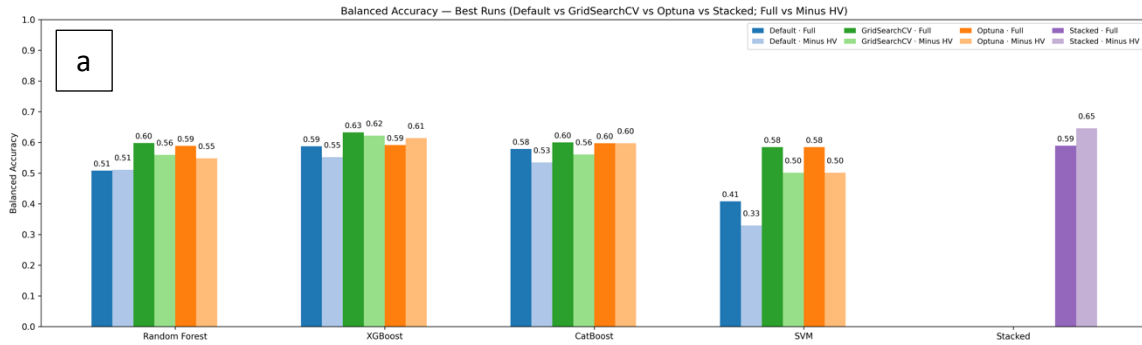


Figure 4. Effects of hyperparameter optimization and stacking on model performance across hypertension subtypes.

(a) Balanced accuracy for each model (Random Forest, XGBoost, CatBoost, SVM) under default settings, GridSearchCV-optimized, Optuna-optimized, and stacked ensemble configurations. The “best runs” represent the hyperparameter sets that achieved the highest minimum sensitivity for each model. Darker bars represent evaluation on the full dataset, while lighter bars represent the patient-only dataset.

(b–e) Sensitivity values for each hypertension subtype, (b) CS, (c) PPGL, (d) PA, (e) PHT, comparing default, optimized, and stacked models. GridSearchCV generally outperformed Optuna, and stacking further improved sensitivity in CS and PA but showed little benefit for PHT and PPGL.

Color key: Blue = Default, Green = GridSearchCV-optimized, Orange = Optuna-optimized, Purple = Stacked ensemble.

3.5 Computational Cost of Optimization and Stacking

Figure 5 summarizes the computational demands and efficiency of hyperparameter optimization and model stacking. While GridSearchCV achieved slightly better performance than Optuna (Fig. 4), it required substantially longer runtimes (Fig. 5a). For Random Forest, GridSearchCV took approximately 1,887 seconds, nearly four times longer than Optuna (474 s), while CatBoost required over 29,700 s (>8 hours) with GridSearchCV, which was nearly twice as long as Optuna (15,212 s). XGBoost showed a smaller but notable difference (3,348 s vs. 2,088 s). Stacking, which combined the best-tuned models, added further computational overhead (7,624 s) despite providing only marginal performance gains over individually optimized models.

Figures 5b and 5c illustrate the search efficiencies of GridSearchCV and Optuna using XGBoost as an example, as similar patterns were observed across the other models. GridSearchCV evaluated a large number of hyperparameter combinations, producing a smooth gradient of balanced accuracy from ~0.56 to >0.64. In contrast, Optuna explored fewer configurations yet achieved comparable performance, reaching ~0.63 with significantly less computational cost.

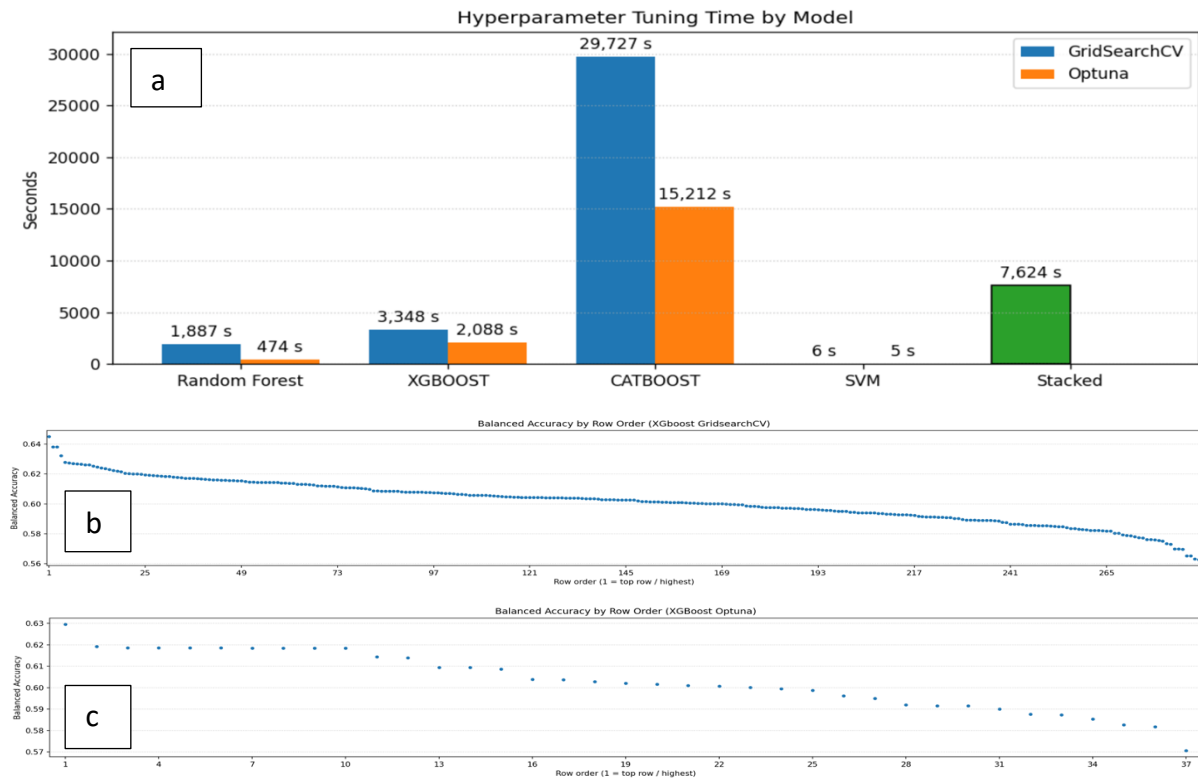


Figure 5. Computational cost and search efficiency of hyperparameter optimization and stacking.

- (a) Total runtime (seconds) required for GridSearchCV, Optuna, and stacking across models. GridSearchCV consistently required substantially longer runtimes than Optuna, particularly for CatBoost (>29,700 s, ~2× Optuna), while stacking introduced additional computational overhead despite marginal performance gains.
- (b) GridSearchCV ranked search results for XGBoost, across hundreds of evaluated configurations. Each dot represents one tested hyperparameter set, ordered from the highest balanced accuracy (left) to the lowest (right).
- (c) Optuna ranked search results for XGBoost, demonstrating comparable performance (~0.63) with significantly fewer evaluated configurations. Each dot represents an evaluated hyperparameter set, similarly ordered from best to worst performance. XGBoost is shown as a representative example, as similar efficiency patterns were observed across models.

3.6 SHAP Analysis Reveals Key microRNA Drivers of CS Subtype Classification

To identify the most influential microRNAs contributing to model classification, SHAP analysis was performed on tree-based models (Random Forest, XGBoost, and CatBoost). Across all models and datasets (full and patient-only), M587063 consistently emerged as the top contributing feature for predicting the CS subtype. Because the dataset uses ΔCt values (inversely proportional to expression), low ΔCt (high expression) of M587063 was associated with positive SHAP values, favoring CS classification, whereas high ΔCt (low expression) opposed CS classification. Within the XGBoost model, M587063 contributed the most

strongly, accounting for approximately 10% of the overall model output in both dataset settings. The XGBoost beeswarm plot is shown as a representative example (Fig. 6a) because similar feature rankings and SHAP distributions were observed across models and datasets. To further validate this finding, ΔCt distributions of M587063 across disease subtypes were examined (Fig. 6b). The CS group displayed the lowest median ΔCt value (0.02), consistent with markedly higher expression compared to other groups.

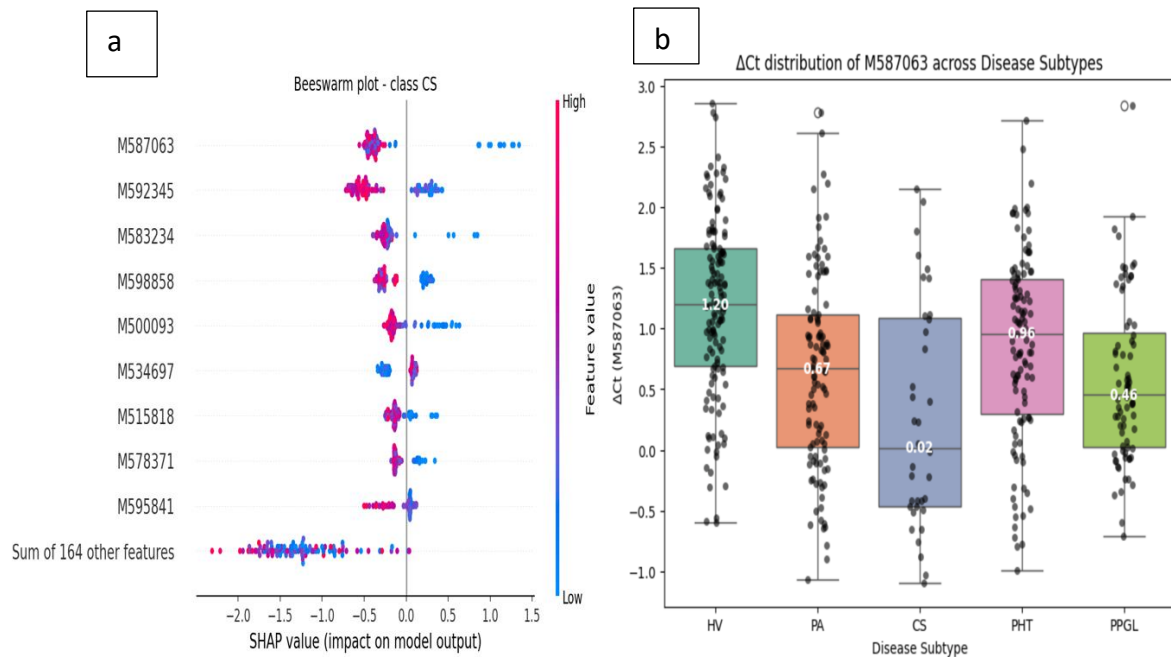


Figure 6. SHAP analysis and expression distribution of M587063 in CS classification.

(a) SHAP beeswarm plot (XGBoost example) summarizing the relative contribution of individual microRNAs to model predictions for the CS subtype. Each dot represents a single sample, positioned horizontally by its SHAP value, which reflects impact on model output. Positive SHAP values favor CS classification, whereas negative values oppose CS classification. Dot color indicates normalized ΔCt value (blue = low ΔCt /high expression; red = high ΔCt /low expression). The y-axis lists the top-ranked microRNA feature identifiers (names blinded to prevent bias). M587063 showed the strongest contribution, with high expression strongly promoting CS classification. Only the XGBoost plot is displayed for clarity, as similar SHAP patterns were observed in Random Forest and CatBoost.

(b) Box plot showing ΔCt distribution of M587063 across disease subtypes, with individual data points overlaid. Median values are indicated. The CS group exhibited the lowest median ΔCt (0.02), corresponding to the highest relative expression compared with other groups.

4.Discussion

Hypertension is a heterogeneous condition encompassing multiple subtypes (CS, PA, PPGL, PHT). Accurate identification of these subtypes is clinically critical, as treatment strategies differ substantially. However, diagnostic workflows remain resource-intensive, and often inconclusive, leading to delayed or missed diagnoses, particularly for rarer forms like CS and PPGL[2,3].

This study evaluated the potential of circulating microRNA profiles, analyzed using machine learning (ML), to classify hypertension subtypes in a clinically meaningful way. Four supervised ML models, Random Forest, XGBoost, CatBoost, and SVM, were benchmarked under default settings, and their performance was iteratively optimized through hyperparameter tuning (GridSearchCV and Optuna) and ensemble stacking.

4.1 Model performance and impact of optimization

Default models demonstrated moderate overall balanced accuracy (0.50) but performed poorly on minority subtypes, with sensitivities close to, and sometimes at, zero for CS and PPGL. This limitation is critical in clinical practice because failing to identify rare but clinically important subtypes means the models will have little diagnostic utility. To address this, the models were retrained with class-weighting to mitigate imbalance, and hyperparameter optimization was performed using GridSearchCV and Optuna independently.

Both tuning approaches improved sensitivity for minority classes, particularly the smallest group, CS, with GridSearchCV providing slightly better results overall. For example, in CS, sensitivity improved from 0.00 in default Random Forest to ~0.42 after tuning and further increased to 0.71 following model stacking. Stacking also yielded marginal gains in overall balanced accuracy (from ~0.63 to ~0.65), suggesting that combining optimized models offered limited additional benefits.

4.2 Trade-offs between improving minority-class sensitivity and majority-class performance

While optimization successfully enhanced detection of CS and PPGL, these gains came at a generally small cost to performance in larger classes such as PHT, where sensitivity slightly worsened across most models. From a clinical perspective, this trade-off may be acceptable if rare but clinically significant subtypes can be better detected; however, overall balanced accuracy remained moderate (~ 0.65), limiting diagnostic utility.

This reflects a broader challenge in ML for imbalanced biomedical datasets: improving minority-class sensitivity often reduces majority-class performance, requiring careful prioritization of optimization objectives[30]. In this study, we deliberately optimized for minimum sensitivity to strengthen the weakest part in the models. While this succeeded in improving detection of small but important subtypes, overall diagnostic utility remained insufficient for clinical deployment.

4.3 Diminishing returns and computational considerations

Although hyperparameter tuning enhanced performance, the computational costs were significant. GridSearchCV was particularly resource-intensive, requiring over 8 hours on the high-performance computer (HPC) for CatBoost and taking nearly 4 times longer than Optuna for Random Forest, while achieving only marginally better results. Stacking further increased computation time ($\sim 7,600$ seconds) but yielded only marginal gains in balanced accuracy.

This suggests that model performance approached a practical ceiling: beyond a certain point, further optimization yields diminishing returns relative to computational investment. Models like XGBoost and CatBoost, which performed well by default, showed limited gains, whereas SVM, the weakest default performer, benefited the most from tuning while requiring fewer computational resources. Future workflows could therefore prioritize computational efficiency by optimising simpler, much faster models like SVM while keeping default settings for stronger, more computationally tree-based models.

4.4 Comparison with previous studies

Our results are broadly comparable to Reel et al. (2025) [6], who applied ML to the same retrospective microRNA dataset and reported a balanced accuracy of 0.73. However, their

metric was computed as the average of sensitivity and specificity, which can be inflated under class imbalance due to the dominance of true negatives in specificity. In contrast, we prioritized macro-recall, the mean sensitivity across subtypes, to ensure that smaller, clinically important classes were not overshadowed by majority groups. When Reel et al.'s results are recalculated using this approach, their balanced accuracy approximates 0.63, closely aligning with our optimized pipeline (0.65).

Additionally, Reel et al. investigated binary classification tasks (e.g., CS vs. PHT), which provide valuable insights for distinguishing specific subtype pairs. Our study instead focused on the multi-class classification problem, where all four subtypes were predicted simultaneously, representing a complementary but inherently more complex approach.

4.5 Insights from multi-omics studies

A separate study by Reel et al. (2022) [9] demonstrated that multi-omics-based classifiers, combining data from multiple biological layers such as proteomics and metabolomics, consistently outperformed mono-omics approaches like ours. Since this study relied solely on circulating microRNA profiles, its predictive power was constrained by the limited biological information contained within a single omics layer. Integrating microRNA data with complementary omics layers could enhance discriminative power by capturing broader mechanistic patterns underlying disease processes.

4.6 Biological insights from SHAP analysis

To better understand model decision-making, SHAP analysis was used to identify microRNAs most influential in classification. M587063 consistently ranked as the top contributing feature across models and datasets, with high expression favoring classification into CS. While the biological identity of M587063 remains blinded to prevent analyst bias, future work should aim to characterize its role in endocrine hypertension. Notably, Vetrivel et al.[31] reported that several microRNAs, including miR-1247-5p, MiR-379-5p, MiR-146-5p, MiR-139-3p, MiR-503-5p, were upregulated in CS, suggesting potential overlap worth investigating once identifiers are revealed.

4.7 Limitations

This study has several limitations. First, analyses relied on a retrospective dataset, which introduced heterogeneity due to differences in sample collection, storage, and processing across sites and time[32]. The use of anonymized data limited the ability to correlate microRNA profiles with clinical characteristics or perform stratified analyses by patient history. Moreover, only microRNA data were available, whereas complementary multi-omics layers could have provided richer biological context and improved classification performance. Another limitation was the absence of independent external datasets, which prevented assessment of model generalizability beyond the training cohort[33]. In addition, the scope of model benchmarking was restricted to four commonly used classifiers due to time and computational constraints, leaving other potentially useful algorithms unexplored. Finally, while SHAP analysis highlighted promising candidate microRNAs such as M587063, their biological identity remained blinded and no experimental validation could be performed within the study timeframe.

4.8 Future Work

Future research should aim to address these limitations. Prospective studies using standardized sample collection and processing protocols are required to reduce technical heterogeneity[32]. Access to non-anonymized datasets with linked clinical metadata would allow integration of demographic and clinical features, improving interpretability and clinical relevance. Expanding analyses to include external validation cohorts is essential for assessing generalizability and real-world diagnostic utility[33]. Integration of additional omics layers, such as transcriptomics, proteomics, and metabolomics, could enhance discriminative power by capturing complementary biological signals[9]. Beyond the current four classifiers, future work could also explore deep learning and other advanced algorithms, provided that sufficiently large datasets become available to support their application[34]. Finally, unblinding and experimental validation of top-ranked miRNAs, such as M587063, will be necessary to confirm their biological roles and assess their potential as clinically actionable biomarkers.

4.9 Conclusion

In this study, circulating microRNA profiles were evaluated for classification of hypertension subtypes using supervised machine learning. Through the use of class weighting, hyperparameter tuning, and stacking, sensitivity for the rarest subtype, CS, was improved from 0.00 under default settings to 0.71. The weakest aspect of model performance, minimum sensitivity, was raised from near zero to approximately 0.50 across models, with the exception of XGBoost, which already performed well at baseline. Nevertheless, overall balanced accuracy remained moderate (~ 0.65) despite these optimizations.

Taken together, these findings demonstrate that although circulating microRNA profiles alone are not yet sufficient for reliable clinical application, optimization of ML models with fairness-oriented evaluation metrics can markedly improve detection of minority subtypes in imbalanced datasets. This provides proof-of-concept that combining miRNA profiling with advanced ML approaches holds promise as a non-invasive diagnostic strategy for hypertension subtypes.

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